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Institutional Quality and Peru's Export Performance: A Comparative Analysis of Panel and Gravity Model Approaches

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Abstract

This study investigates the association between institutional quality — measured through the World Bank's Worldwide Governance Indicators (WGI) — and Peru's export performance over the 2015–2024 period. The analysis applies two frameworks to the same trade data: (i) a bilateral panel regression on 12 partners (the other ten original CPTPP parties, the United Kingdom, and China), and (ii) an OLS gravity model (Tinbergen 1962) of Peru's exports to 184 partners. A shift-share decomposition (Tyszynski 1951; Leamer and Stern 1970) separates Peru's export growth from US37.4 billion (2015) to US76.8 billion (2024) into a world-demand component (40.1%), a structural-composition component (19.0%), and a competitiveness residual (40.9%). The panel regression finds no statistically significant association between the partner's lagged governance and that competitiveness residual (within-R^2 = 0.002, $p > 0.05$). The gravity model, with 1,496 observations, estimates a coefficient of 0.667 ($p < 0.001$, standard errors clustered by partner) on the destination's lagged WGI. Peru is the only exporter, so its own WGI does not vary across partners within a year and is absorbed by the year fixed effects. The coefficient is an association between destination governance and Peruvian exports to that destination. It is not an estimate of the effect of Peru's institutions. The association survives a two-year lag (coefficient 0.668, $p < 0.001$) and the outlier and aggregation checks that were estimated. It does not survive first differences ($p = 0.906$). A path from current exports to subsequent governance is significant ($p < 0.001$). Poisson pseudo-maximum likelihood was not estimated, so the paper does not claim robustness across estimators. All six destination-governance dimensions are individually significant in separate regressions. No causal claim is made.

Keywords: institutional governance, Worldwide Governance Indicators, export competitiveness, bilateral gravity model, shift-share decomposition, Peru, CPTPP.

JEL Classification: F10, F13, F14, O17, O24.

1. Introduction

The hypothesis that better institutions drive higher trade volumes has been a central proposition in development economics since the seminal work of North (1990) on transaction costs and the institutional foundations of markets. Countries with stronger rule of law, more effective governments, and lower corruption are expected to incur lower transaction costs in cross-border commerce, attract higher-quality foreign direct investment, and sustain more predictable commercial relationships — all of which should translate into larger export volumes. Despite a growing empirical literature, however, findings remain sensitive to methodological choices: sample composition, model specification, measurement of institutional quality, and the level of product aggregation all affect whether a governance effect is detected.

This paper examines the specific case of Peru, an open, commodity-dependent economy in Latin America that has signed multiple free trade agreements including the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP). Peru's experience is instructive because it combines moderate movement in some governance dimensions over 2015–2024 with a near-doubling of its export base. The study's primary contribution is methodological: it applies two empirical frameworks and shows that they answer different questions. The panel asks whether year-to-year changes in a partner's governance line up with Peru's competitiveness residual in that market. The gravity model asks whether Peru exports more to destinations with higher governance scores. Neither design identifies the effect of Peru's own institutions on its exports. A null result in the 12-partner panel does not refute a governance hypothesis, and a significant gravity coefficient does not establish one for Peru.

The paper is structured as follows. Section 2 reviews the relevant literature. Section 3 presents the theoretical framework linking each WGI dimension to trade costs. Section 4 describes the data sources and construction. Section 5 details the three methodological components: shift-share decomposition, bilateral panel regression, and the bilateral gravity model. Section 6 reports the results for each. Section 7 discusses the reasons for the divergent findings. Section 8 acknowledges limitations. Section 9 states what the estimates can and cannot inform. Section 10 concludes.

2. Literature Review

2.1 Governance and International Trade

The empirical literature on institutional quality and trade flows spans multiple decades and methodologies. Wei (2000) studies foreign direct investment, not merchandise trade. In a sample of twelve source countries and 45 host countries, a rise in host-country corruption is associated with lower inward FDI; in his benchmark, moving from Singapore's corruption level to Mexico's has a negative association comparable to a large increase in the corporate

tax rate. Anderson and Marcouiller (2002) estimate a structural model of import demand in which insecurity acts as a hidden tax on trade. Their abstract reports that inadequate institutions constrain trade by about as much as tariffs.

Álvarez, Barbero, Rodríguez-Pose, and Zofío (2018) estimate a sectoral gravity equation for 186 countries over 1996–2012. Institutional conditions at the destination and institutional distance between partners are associated with bilateral trade; the destination association is larger for agriculture and raw materials than for manufactures and services, and it rises moderately over time. The published summary of de Groot, Linders, Rietveld, and Subramanian (2004) reports that a similar institutional framework is associated with about 13% more bilateral trade, and that a one-standard-deviation increase in institutional quality is associated with about 30–44% more bilateral trade, depending on whether the change is on the importer or the exporter side. Those percentages are not the 29–56% range cited in an earlier draft of this paper. Méon and Sekkat (2004) find that institutional quality is associated with manufactured exports and with FDI attractiveness in Middle East and North Africa countries during the 1990s. It does not identify an informal-corruption channel as distinct from a tariff-equivalent channel.

2.2 Gravity Models and Institutional Quality

The bilateral gravity model of trade, originally proposed by Tinbergen (1962) and theoretically grounded by Anderson (1979) and Anderson and van Wincoop (2003), has become the standard framework for estimating the association between institutional quality and trade. The gravity equation predicts bilateral trade as a function of the economic masses of trading partners (typically GDP) and bilateral resistance factors (distance, borders, language). Institutional quality enters as an additional correlate of trade costs, alongside geography and policy barriers.

Abreo, Bustillo, and Rodriguez (2021) estimate a gravity model of Colombian exports to 136 partners over 2005–2018. Colombian institutional quality and the institutional distance from partners are associated with export performance, and the authors highlight regulatory quality and the rule of law. Their specification includes the exporter's own institutions, so it is not the same object as the destination-only coefficient estimated below. Riker (2022) estimates the association between a country's WGI regulatory-quality score and its GDP and its manufactured trade with the United States. The trade associations are larger, in percentage terms, for developing countries. The working paper does not report the bilateral sample or the minimum panel size that an earlier draft attributed to it. Those claims are withdrawn.

2.3 The Latin American Context

For Latin American countries, the directly relevant evidence is thinner than an earlier draft of this paper stated. Volpe Martincus and Carballo (2008) use firm-level data for Peru over 2001–2005 and find that export-promotion activities are associated with higher exports,

mainly along the extensive margin of products and destinations. The paper studies promotion by PROMPEX. It does not measure customs efficiency, and it is not a multi-country panel of WGI scores. Three citations used in the earlier draft do not exist in the form in which they were given and are withdrawn. A fourth attribution, linking institutions to export sophistication, is also withdrawn: the book that was cited does not report that result and is not used here. The withdrawn titles are listed in the verification note, not in the bibliography.

2.4 Gaps in the Literature

Three gaps motivate this study. First, most existing studies use a single methodological framework; the practice of contrasting a panel regression with a gravity model on the same data is rare. Second, Latin American case studies using the WGI at the bilateral level are relatively few. Third, null findings in governance-trade studies are often not published, creating a publication bias toward positive results. This paper addresses all three gaps by documenting the full research trajectory — including the initial null finding — as an explicit contribution to methodological transparency.

3. Theoretical Framework

3.1 Institutions and Trade Costs

The canonical gravity model of Anderson and van Wincoop (2003) derives bilateral trade flows from a CES expenditure system in which the resistance terms — the "iceberg" costs of moving goods between countries — determine the volume of trade. Institutional quality enters this framework by increasing the effective trade resistance: weak institutions raise the costs of enforcing contracts, clearing customs, and securing property rights, all of which act as non-tariff barriers. Formally, if τ_{ij} represents the bilateral trade cost between i and j, and G_i and G_j represent the institutional quality of each country, then $\tau_{ij} = f(G_i, G_j, d_{ij})$, where d_{ij} is geographic distance. Better institutions reduce τ_{ij} and increase trade flows.

This theoretical grounding implies that institutional quality should have a positive effect on trade flows in a gravity framework, holding income and distance constant. The effect should be heterogeneous across product categories, since goods requiring contract enforcement (complex manufactures, services exports) are more institution-sensitive than standardized commodities.

3.2 The Six WGI Dimensions and Their Trade Channels

The World Bank's Worldwide Governance Indicators (WGI) measure six dimensions of governance. Each dimension maps to a distinct mechanism affecting trade costs:

Control of Corruption (CC): Reflects the extent to which public power is exercised for private gain, including both petty bribery and grand corruption. For exporters, corruption raises effective trade costs through informal payments at customs, preferential treatment for connected firms, and reduced predictability of regulatory outcomes. A country with a CC score one standard deviation above the mean can expect lower per-shipment informal costs and shorter customs clearance times, both of which improve competitiveness.

Government Effectiveness (GE): Captures the quality of public service provision, the competence of civil servants, and the degree of policy credibility. Effective governments produce clearer and more consistent regulations, faster administrative procedures, and lower uncertainty for commercial operators. For exporters, GE affects the time and cost of obtaining export licenses, sanitary certificates, and customs clearances — all direct inputs to the trade cost function.

Rule of Law (RL): Measures the extent to which agents have confidence in and abide by the rules of society, including the quality of contract enforcement, property rights, and judicial independence. Strong rule of law reduces the risk that foreign buyers default on payment obligations and increases the willingness of exporters to extend credit to buyers in that country. This is particularly relevant for complex manufactured goods and capital equipment.

Regulatory Quality (RQ): Captures the ability of the government to promote private sector development through sound regulations and policies. Excessive or unpredictable regulation raises compliance costs for exporters, particularly in sectors subject to sanitary and phytosanitary standards, technical regulations, and product certification requirements. Better regulatory quality reduces these costs and expands the set of viable export markets.

Political Stability and Absence of Violence/Terrorism (PS): Measures the likelihood that the government will be destabilized by unconstitutional or violent means. Political instability raises the risk premium on commercial transactions, increases insurance costs, and may disrupt physical supply chains. For Peru's mining and agricultural exporters — sectors sensitive to infrastructure access — political instability in destination markets directly affects logistics costs and delivery reliability.

Voice and Accountability (VA): Captures the extent to which citizens can participate in the selection of government and enjoy freedoms of expression, association, and media. While VA is the most indirect of the six dimensions for trade costs, it serves as an indicator of policy stability and predictability: governments subject to greater accountability are less likely to make sudden, protectionist policy changes that disrupt established trade relationships.

3.3 The Shift-Share Decomposition as a Competitiveness Filter

The shift-share framework (Tyszynski 1951; Leamer and Stern 1970) decomposes export growth into three additive components:

$$\Delta X_{peru} = \Sigma_c [X_{peru,c,0} \cdot g_{world}] + \Sigma_c [X_{peru,c,0} \cdot (g_{world,c} - g_{world})] + \Sigma_c [X_{peru,c,1} - X_{peru,c,0} \cdot (1 + g_{world,c})]$$

where the first term is the world demand effect (what Peru would have earned if each chapter grew at the world average rate), the second term is the structural composition effect (the gain or loss from selling in chapters growing faster or slower than the world average), and the third term is the residual — the genuine competitiveness effect — which is what cannot be explained by world growth rates alone. This third component is the variable of interest in the bilateral panel regression: it captures whether Peru gained or lost market share in each partner country, year by year, at the HS2 chapter level.

4. Data

4.1 World Governance Indicators (WGI)

The WGI are produced by the World Bank and aggregate data from 31 underlying data sources covering over 200 countries and territories from 1996 to the present. Each indicator is measured on a standardized scale from approximately –2.5 (weakest governance) to +2.5 (strongest governance) and is available as a point estimate and as a margin of error. For this study, the six dimensions were retrieved for the period 2015–2024 for all countries appearing in the trade datasets. The WGI for each country-year pair were averaged across the six dimensions to produce a composite governance score (WGI_avg), following the aggregation approach used in the literature (e.g., Kaufmann, Kraay, and Mastruzzi 2011). All WGI values were lagged by one year (t–1) relative to the dependent trade variable to reduce simultaneity concerns.

4.2 Trade Data: BACI (CEPII)

The primary trade data source is BACI (Base Amiable du Commerce International), version 202601, published by the CEPII (Centre d'Études Prospectives et d'Informations Internationales). BACI provides harmonized bilateral trade flows at the HS6 product level, reconciled between reporter and partner country customs data to minimize measurement error. For this study, flows are aggregated to the HS2 chapter level to reduce noise while maintaining sufficient product-level differentiation. Two datasets were constructed from BACI: (i) Peru's exports to the world by chapter and year (2015–2024), and (ii) Peru's exports to each individual trade partner by chapter and year, covering 212 partner countries. All values are in FOB (Free on Board) thousands of US dollars.

For the gravity model, the bilateral panel uses Peru's total exports to each partner country per year, yielding 1,837 observations at the raw level, reduced to 1,496 after merging with WGI, GDP, and geographic controls.

4.3 Geographic Controls: GeoDist (CEPII)

Bilateral geographic controls — distance (in km), contiguity (shared land border), shared official language, and colonial relationship — are drawn from the CEPII GeoDist database (Mayer and Zignago 2011). Distance is measured as population-weighted bilateral distance. These variables are time-invariant at the country-pair level and are the standard gravity controls in the trade literature.

4.4 GDP Data

Partner country GDP in current US dollars is obtained from the World Bank World Development Indicators (WDI), matching the WGI dataset by country-year. GDP is log-transformed in the gravity model, following standard practice.

4.5 Descriptive Statistics

Table 1 presents descriptive statistics for the key variables in the gravity model (1,496 observations, 184 countries, 2016–2024).

Table 1: Descriptive statistics, bilateral gravity model, 1,496 observations.

Variable	Mean	Std. Dev.	Min	Max
log(Exports, thousands USD)	6.85	2.41	–4.61	15.19
log(GDP, current USD)	25.02	1.98	21.10	30.21
log(Distance, km)	8.88	0.87	6.57	9.87
WGI Average (t–1)	0.06	0.88	–1.95	1.89
Contiguity	0.03	0.17	0	1
Shared Official Language	0.11	0.31	0	1
Colonial Relationship	0.03	0.17	0	1

5. Methodology

5.1 Shift-Share Decomposition

The shift-share decomposition is applied to Peru's total exports (2015–2024) at the HS2 chapter level using the world as the reference market. For each chapter c, the growth rate of world exports of that chapter (g_world,c) is calculated between 2015 and 2024. The aggregate world export growth rate (g_world) is the growth of total world exports, which equals an export-weighted average of the chapter growth rates, not a production-weighted average. The three components are summed across all 96 chapters present in the Peru dataset. The accounting identity requires those three components to add up to the observed change in Peru's exports. It does not require the competitiveness residual to have a mean of zero. In the estimates below, that residual is US16.1 billion, 40.9% of the change. The residual of the identity, reported in Table 2, is 0.02 thousand US dollars (about US20), which is inside a tolerance of US$1,000.

5.2 Bilateral Panel Regression (CPTPP+China)

The bilateral panel regression uses 12 partner countries over 2016–2024. Peru is the exporter and is not one of the 12. The partners, taken from the replication scripts, are Australia, Brunei Darussalam, Canada, Chile, China, Japan, Malaysia, Mexico, New Zealand, Singapore, Vietnam, and the United Kingdom. That list is the other ten original CPTPP parties, plus the United Kingdom, plus China. It is not "the 11 original CPTPP members plus China": the original signatories include Peru, and the United Kingdom was not an original signatory. For each partner-year, the bilateral shift-share competitiveness component is calculated using the partner's import market as the reference instead of the world market. The dependent variable is that competitiveness component for Peru's exports to the partner in that year, in thousands of US dollars.

The regression model takes the form:

$$Y_{jt} = \alpha + \beta \cdot WGI_{j,t-1} + X_{jt}' \cdot \gamma + \mu_j + \lambda_t + \varepsilon_{jt}$$

where Y_{jt} is the competitiveness effect in partner j in year t, $WGI_{j,t-1}$ is the average WGI score for partner j, lagged one year, X_{jt} is a vector of time-varying controls (partner GDP, exchange rate), μ_j are partner fixed effects, λ_t are year fixed effects, and ε_{jt} is the error term. The model is estimated by fixed-effects panel OLS with clustered standard errors at the partner-country level. An equivalent random-effects model is also estimated, and the two are compared.

The 12-partner subsample is of particular interest because it reflects Peru's most strategically important trade relationships — all CPTPP members plus China account for a substantial share of Peru's total exports — and because it corresponds to the policy-relevant question: does governance quality in Peru's preferential trade partners affect its competitive position in those markets?

5.3 Bilateral Gravity Model

The standard bilateral gravity model follows Tinbergen (1962) as extended by Anderson and van Wincoop (2003):

$$\ln(X_{jt}) = \beta_0 + \beta_1 \cdot \ln(GDP_{jt}) + \beta_2 \cdot \ln(DIST_j) + \beta_3 \cdot WGI_{j,t-1} + \beta_4 \cdot CONTIG_j + \beta_5 \cdot COMLANG_j + \beta_6 \cdot COLONY_j + \lambda_t + \varepsilon_{jt}$$

where X_{jt} is Peru's exports to partner j in year t (thousands of US dollars), GDP_{jt} is partner j's GDP in year t (current US dollars), $DIST_j$ is bilateral distance, $WGI_{j,t-1}$ is the lagged average WGI score of the destination, $CONTIG_j$, $COMLANG_j$, and $COLONY_j$ are time-invariant geographic controls, and λ_t are year fixed effects. The model is estimated by OLS with standard errors clustered by partner. Clustering allows arbitrary correlation within a partner over time. Cameron, Gelbach, and Miller (2011) study the multiway extension of this estimator; the estimates here use one-way clustering only. The year fixed effects absorb shocks common to all partners, including Peru's own WGI, which does not vary across destinations in a given year. The coefficient on $WGI_{j,t-1}$ is therefore not the effect of Peruvian institutions.

The gravity model is estimated over 184 partner countries (representing Peru's complete set of bilateral trade relationships with non-zero flows) and 9 years (2016–2024), yielding 1,496 observations.

5.4 Endogeneity and Robustness

Three strategies are used to assess the robustness of the gravity model results and to probe potential endogeneity:

(a) Lag structure comparison: The lag is extended from t–1 to t–2, following the logic that a causal effect of governance on trade should survive a longer lag, while a contemporaneous correlation driven by simultaneity should weaken or disappear.

(b) Estimator not reported. Santos Silva and Tenreyro (2006) show that OLS on log exports drops zero flows and can be biased under heteroskedasticity, and they propose Poisson pseudo-maximum likelihood. This version does not estimate PPML. Section 4.2 records 212 partners in the BACI extract and 1,837 raw partner-year observations, against 184 partners and 1,496 observations in the OLS sample. The paper does not split that loss into zero flows and missing WGI, GDP, or geographic data. Robustness to the estimator is not a result of this paper.

(c) Subsample analysis: The gravity model is re-estimated on the CPTPP+China subsample (12 countries) to test whether the panel result is recoverable within the gravity framework when the same small sample is used.

6. Results

6.1 Shift-Share: Decomposition of Peru's Export Growth 2015–2024

Table 2 presents the shift-share decomposition of Peru's export growth between 2015 and 2024.

Table 2: Shift-share decomposition of Peru's export growth, 2015–2024, BACI (CEPII).

Component	Thousand USD	% of Growth
World Demand Effect	15,809,655	40.1%
Structural Composition Effect	7,488,565	19.0%
Real Competitiveness Effect	16,128,987	40.9%
Total Observed Growth	39,427,207	100%
Discrepancy (rounding)	0.02	—

Peru's exports grew from US37.4 billion in 2015 to US76.8 billion in 2024 (+US$39.4 billion, +105.4%). Of this growth, 40.1% is attributable to favorable world demand conditions — Peru benefited from an expanding global market, particularly for minerals and agricultural commodities. The structural composition effect (19.0%) reflects that Peru's export bundle was concentrated in chapters that grew faster than the world average (mining products, certain agricultural products), a favorable "structural luck" component. The real competitiveness effect (40.9%) represents the component attributable to Peru gaining market share within each chapter, over and above what world growth would predict. This residual is the dependent variable in the bilateral panel regression.

The three components sum to the observed growth up to the residual in Table 2, which is 0.02 thousand US dollars (about US20). Section 5.1's tolerance of less than US1,000 refers to that same residual. The competitiveness component itself is US$16.1 billion. It is not a zero-mean residual.

6.2 Bilateral Panel Regression: CPTPP+China

Table 3 presents the results of the bilateral panel regression for the CPTPP+China subsample.

Table 3: Bilateral panel regression results, 12 CPTPP+China partners, 108 observations (2016–2024).

Specification	WGI Coefficient	Std. Error	t-statistic	p-value	R^2 (within)
Fixed Effects (Entity + Time)	–125,700	636,500	–0.197	0.844	0.002

Random Effects	3,201	58,100	0.055	0.956	0.00003
Variant A: Dep. var normalized	–0.0021	0.0010	–2.100	0.051	—
Variant A: Without Brunei	–0.0020	0.0011	–1.797	0.076	—
Variant A: No Time Effects	–0.0028	0.0011	–2.702	0.008	—

The main regression (columns 1–2) finds a negative but statistically insignificant coefficient on the lagged WGI variable in both fixed and random effects specifications. The within-R^2 of 0.002 indicates that lagged governance explains essentially none of the variation in bilateral competitiveness effects at this level of aggregation. The normalized specifications (dependent variable scaled by the partner's total imports, which eliminates scale effects from large vs. small partner countries) reveal a marginally significant negative association ($p = 0.051$ and $p = 0.008$ depending on the specification), a finding that is directionally counterintuitive (better governance associated with worse competitiveness) and does not survive endogeneity testing (see Section 6.4).

6.3 Bilateral Gravity Model: 184 Partners

Table 4 presents the main gravity model results.

Table 4: Bilateral gravity model — Peru vs. 184 trade partners, 2016–2024. OLS with cluster-robust standard errors by country.

Variable	Coefficient	Robust Std. Error	z-statistic	p-value
Intercept	–2.557	2.373	–1.077	0.281
log(GDP partner)	1.246	0.066	18.755	< 0.001
log(Distance)	–2.164	0.261	–8.274	< 0.001
WGI Average (t–1)	0.667	0.124	5.398	< 0.001
Contiguity	–0.039	0.430	–0.090	0.929
Shared Official Language	0.140	0.361	0.389	0.698
Colonial Relationship	1.186	0.372	3.189	0.001
Year Fixed Effects	Yes	—	—	—
R^2	0.721	—	—	—

Observations	1,496	—	—	—
Countries	184	—	—	—
Years	9	—	—	—

The gravity model produces a result sharply different from the panel regression. The coefficient on destination WGI is positive and statistically significant (0.667, $p < 0.001$). A 1% higher partner GDP is associated with 1.25% higher Peruvian exports. A 1% greater distance is associated with 2.16% lower exports. The colonial-tie coefficient of 1.186 corresponds to a factor of about 3.27 (the exponential of 1.186). The R^2 of 0.721 is high for this cross-destination regression. These are associations. The specification has no importer-year multilateral-resistance terms, no trade-agreement or tariff controls, and no PPML estimates.

What the WGI coefficient measures. The dependent variable is in logs, so a change of Δ in destination WGI is associated with a percent change of $100 \cdot (\exp(0.667 \cdot \Delta) - 1)$ in Peru's exports to that destination. A one-standard-deviation higher destination score — 0.88 points in this sample — is associated with about 80% higher Peruvian exports to that destination ($\exp(0.667 \times 0.88) - 1 = 0.796$). This is not the effect of raising Peru's WGI. An earlier draft illustrated the coefficient by moving Peru's 2023 score (–0.28) to Chile's (0.78) and reporting a 104% increase in exports to Chile. That comparison applies a destination coefficient to the exporter's score. It is withdrawn. The published summary of de Groot et al. (2004) reports 30–44% for a one-standard-deviation change in a model where institutions vary on both sides of the pair. That object is not the coefficient in Table 4, so the two percentages are not a robustness check of each other.

6.4 Endogeneity and Robustness

Table 5 summarizes the robustness and endogeneity checks on the gravity model.

Table 5: Robustness and endogeneity checks. First-differences and reverse-precedence estimates from the auxiliary script; standard errors clustered by country throughout.

Check	Specification	WGI Coefficient	p-value	Obs.	Result
Baseline (t–1 lag)	Log-exports, OLS cluster	0.667	< 0.001	1,496	Baseline
t–2 lag	Same model, lag extended	0.668	< 0.001	1,325	Survives
First differences	Δlog(export) ~ ΔWGI + Δlog(GDP)	–0.063	0.906	1,312	Does not survive
Reverse precedence	WGI(t+1) ~ log(export) t + controls	—	< 0.001	1,339	Significant reverse path

Excluding outliers (top/bottom 1% WGI)	Trimmed sample	0.682	< 0.001	1,466	Survives
Median WGI (instead of mean)	Aggregate with median	0.613	< 0.01	1,496	Survives
CPTPP+China subsample (N=12)	Same model, N=12	0.412	0.080	108	Directionally consistent, not significant
Panel regression t–2 lag	Fixed effects	–0.0001	0.896	96	Effect disappears in panel

The t–2 lag test is the most informative of the lag-based checks: extending the lag by one year produces a coefficient of 0.668, virtually identical to the baseline (0.667). This is the same test that, when applied to the panel regression with 12 countries, caused the effect to disappear entirely (coef = –0.0001, p = 0.896). The contrast is telling. In the gravity model with 184 countries, there is sufficient variation in WGI across both countries and time to identify the effect even at a two-year lag. In the panel with 12 countries, the variation is exhausted after one year, and the second-lag test reflects only noise.

Two further checks qualify the interpretation, and we report them because omitting them would overstate the evidence. **First differences** — regressing year-on-year changes in log exports on year-on-year changes in lagged WGI, which removes all time-invariant country heterogeneity — shrink the coefficient to –0.063 and render it insignificant (p = 0.906). **Reverse precedence** — regressing WGI(t+1) on current log exports with the same controls — yields a significant positive coefficient (p < 0.001), indicating that export performance predicts subsequent governance scores as well. Taken together, these results are consistent with the estimating variation being predominantly between-country rather than within-country: the association lives in levels and across countries, not in changes over time within a country.

The honest summary is therefore that the governance–export association in the gravity model is **robust as a levels relationship** — robust to lag extension, outlier removal, and aggregation choices — but that the design does not support a causal claim. The endogeneity battery produces the same verdict that the literature reports for comparable exercises: the association is stable and replicable, while causal identification remains out of reach without a valid instrument.

The CPTPP+China subsample of the gravity model yields a coefficient of 0.412 (p = 0.080), directionally consistent with the full-sample result but not statistically significant at conventional levels — reflecting the loss of power when reducing from 184 to 12 countries.

6.5 Dimension-Level Estimates

Table 6 reports the gravity model estimated separately for each of the six WGI dimensions, replacing the composite score one dimension at a time.

Table 6: Gravity model with each WGI dimension entered separately (N = 1,479; 182 countries). The reduction from 1,496 observations reflects countries missing at least one dimension.

WGI Dimension	Coefficient	p-value	Ranking
Regulatory Quality	0.681	< 0.001	1
Government Effectiveness	0.671	< 0.001	2
Rule of Law	0.599	< 0.001	3
Political Stability and Absence of Violence	0.580	< 0.001	4
Voice and Accountability	0.579	< 0.001	5
Control of Corruption	0.552	< 0.001	6

All six dimensions are individually positive and statistically significant, and the spread across them is narrow: the strongest coefficient exceeds the weakest by only 0.13 points. This should temper any strong claim about a single dominant dimension. We note two caveats on interpretation. First, when all six dimensions are entered simultaneously, none is individually significant — a mechanical consequence of the pairwise correlations among WGI dimensions, which range from roughly 0.7 to 0.9. The separate-entry specification in Table 6 is therefore the informative one; the joint specification is reported for completeness and not as a substantive model. Second, the ranking should be read as suggestive rather than definitive, given the narrow spread and the collinearity. What the table establishes is that the association between destination governance and Peruvian exports is not driven by any single WGI dimension. Regulatory Quality and Government Effectiveness have the largest point estimates. That ranking describes destinations. It is not a sequence for reforming Peruvian institutions.

7. Discussion

7.1 Why Does the Panel Miss the Effect?

The useful contrast is between two designs, not between two estimates of the same effect. Abreo, Bustillo, and Rodriguez (2021) and de Groot et al. (2004) include institutional quality on the exporting side, or on both sides. Table 4 does not.

Statistical power. The panel has 12 clusters and 9 years. Partner fixed effects and year fixed effects use up degrees of freedom, so "108 degrees of freedom" overstates what is left for the WGI slope. Clustered standard errors with 12 clusters are a weak basis for

inference. An earlier draft reported a post-hoc power of 18% for an effect of "0.667 SD units," set against 99% power in the gravity sample. That calculation compares a log-point semi-elasticity from one dependent variable with a residual measured in thousands of US dollars in another. It is withdrawn. A null result in this panel is not evidence that destination governance is irrelevant, and it is not evidence about Peru's own institutions.

The competitiveness residual is not zero-mean. The panel's dependent variable is the shift-share competitiveness component: actual export growth minus the growth implied by the reference market's chapter growth rates. The accounting identity requires the demand, composition, and competitiveness components to sum to the total change. In the aggregate decomposition they sum to US39.4 billion, of which competitiveness is US16.1 billion. Nothing in that identity forces the competitiveness term to have a mean of zero, and Table 2 shows that it does not. The gravity model instead uses the log level of exports, which varies strongly across destinations with GDP and distance. The two dependent variables are not rescalings of each other.

Entity fixed effects absorbing variation. The panel's entity fixed effects (μ_j) absorb all time-invariant country heterogeneity, including the part of WGI that is fixed over the 9-year period. For most countries, the WGI score changes slowly over 9 years — particularly for high-income countries — which means that the within-entity variation in WGI is small relative to the between-entity variation. The gravity model's country-level WGI is predominantly between-country variation, which is exactly what the model exploits.

7.2 Consistency with the Literature

The gravity result (0.667, $p < 0.001$) has the same sign as the destination-institution associations in Álvarez et al. (2018) and as the institutional-quality associations in Abreo, Bustillo, and Rodriguez (2021) and de Groot et al. (2004). It does not match their magnitude, because those papers do not estimate a single-exporter OLS coefficient on destination WGI alone. The published summary of de Groot et al. (2004) reports 30–44% for a one-standard-deviation change, not 29–56%. The figure of about 80% in this paper is $\exp(0.667 \times 0.88) - 1$ for a destination score. It is not a replication of de Groot et al.

7.3 What the Endogeneity Tests Do and Do Not Establish

The endogeneity battery in Table 5 yields a more nuanced verdict than the lag tests alone would suggest, and we state it plainly. The association **survives** the extension of the lag to two years — the same test that annihilates the effect in the 12-country panel. It **does not survive** first differences, and there is a **significant reverse path** from current exports to subsequent governance.

Three readings of this pattern are consistent with the evidence. The first is that the true relationship is a long-run levels relationship: institutions and trade co-determine each other slowly, and year-on-year changes are dominated by noise, transitory shocks, and

measurement error in both WGI and trade data — so first differences discard signal along with bias. The second is that reverse causality is real and substantial: countries that trade more may, over time, invest in the administrative capacity that governance indicators measure, which would make the coefficient partly reflect exports→governance rather than governance→exports. The third is that an unobserved country-level factor — human capital, state capacity, geographic endowment beyond distance — drives both, generating a stable between-country correlation with no causal content.

The data cannot adjudicate among these three. What the data do establish is that the association is not an artifact of the particular lag choice, not driven by outliers, and not specific to one governance dimension. Equally, they establish that the association is not within-country dynamic in character. A study reporting only the levels result would overstate the evidence; a study reporting only the first-differences result would understate it. We report both.

Separating these readings would require a variable that shifts governance and affects trade only through that channel. Acemoglu and Johnson (2005) unbundle property-rights institutions from contracting institutions and show that the historical instruments commonly used for "institutions" do not isolate a single channel. They do not document that no instrument for the governance–trade link exists. This paper does not offer one. The contribution is the association between destination WGI and Peruvian exports, and a record of which checks that association survives.

8. Limitations

Six limitations should be considered when interpreting these results.

The gravity coefficient is not Peru's institutions. WGI_{j,t-1} is the destination's score. With one exporter, Peru's score is collinear with the year fixed effects. The regression cannot answer whether institutional reform in Peru raises Peruvian exports. It can describe which destinations receive more Peruvian exports.

Cross-sectional variance in WGI. The WGI indicators have a predominantly cross-sectional (between-country) variance structure. With only 10 years of data, the within-country (over-time) variation is relatively small, which constrains the ability to use WGI changes as a time-varying treatment. Our lagged specification exploits the small within-country variation over time, but most of the identifying variation in the gravity model comes from between-country differences.

Perception-based measurement. The WGI are constructed from expert perception surveys and commercial risk ratings, which may reflect reporting biases rather than objective institutional quality. Countries with more transparent media environments may receive more favorable governance scores independently of the actual quality of their institutions. Nevertheless, the WGI have been validated against objective measures (e.g., firm survey data, objective corruption indices) with generally high correlations.

Omitted variables. While the gravity model controls for GDP, distance, borders, language, and colonial ties, it does not control for trade policy variables such as tariff rates, non-tariff measures, participation in preferential trade agreements (PTAs), exchange rate volatility, or logistics infrastructure. These omitted factors could be correlated with both governance and trade, biasing the WGI coefficient. Peru's multiple PTAs (with CPTPP, the EU, China, the US, and others) are particularly important unobservables.

Product aggregation. The shift-share decomposition and the bilateral panel regression use HS2 chapter-level data (96 chapters). This level of aggregation is standard for macroeconomic trade analysis but obscures product-level heterogeneity. Institution-sensitive products (pharmaceuticals, electronics, services) are averaged together with commodity exports (copper, fishmeal, coffee) where governance effects may be minimal.

Sample period. The 2015–2024 period includes two major shocks — the US-China trade war and the COVID-19 pandemic — that affected Peru's trade patterns in ways that may not be fully captured by year fixed effects. The pandemic in particular caused a temporary collapse in services exports and a surge in commodity exports, which may have affected the composition of Peru's export bundle in ways that interact with governance effects.

9. What the Estimates Can and Cannot Inform

The estimates do not identify a return to institutional reform in Peru, and they do not rank domestic reforms. Section 9 of an earlier draft drew recommendations for customs administration, the Ministry of Economy and Finance, and governance chapters of trade agreements from the coefficient 0.667. Those recommendations are withdrawn. CPTPP Chapter 16 is Competition Policy, Chapter 23 is Development, Chapter 25 is Regulatory Coherence, and Chapter 26 is Transparency and Anti-Corruption. Citing Chapters 26 and 23 as "Transparency" and "Competition Policy" was incorrect, and the policy paragraph that used those citations has been removed rather than repaired.

What Table 4 supports is narrower. Peru's exports are larger, in this OLS sample, toward destinations with higher lagged WGI, conditional on the destination's GDP, distance, contiguity, common language, colonial ties, and year effects. A one-standard-deviation difference in destination WGI (0.88 points) lines up with about 80% higher exports to that destination. Partner GDP (coefficient 1.246) and the colonial-tie coefficient (1.186) are larger in the same regression. The pattern is consistent with market selection: shipments are greater where the destination's measured governance is higher. It is also consistent with omitted destination characteristics that move with WGI, including income per capita, which is not controlled for separately from total GDP, and preferential access, which is not controlled for at all. The 80% figure is not a comparison with a preferential trade agreement.

Table 6 says the same association appears for each destination dimension entered alone. Regulatory Quality (0.681) and Government Effectiveness (0.671) have the largest coefficients, then Rule of Law (0.599), Political Stability (0.580), Voice and Accountability

(0.579), and Control of Corruption (0.552). The spread is 0.13. The dimensions are collinear. The order is not a mandate for which Peruvian agency to reform first. Control of Corruption is the smallest of the six destination coefficients. It does not, by itself, imply a customs or fiscal recommendation.

10. Conclusions

This paper documents the relationship between institutional quality and Peru's export performance using two complementary methodological approaches and finds that the answer depends critically on research design.

Finding 1 (panel): In a bilateral panel with 12 partners and 9 years (108 observations), the partner's lagged WGI has no statistically significant association with Peru's bilateral competitiveness component (within-R^2 = 0.002, $p > 0.05$). With 12 clusters, partner and year fixed effects, and a dependent variable that is not the log of exports, this null does not measure the power of the gravity model and does not say anything about Peru's own WGI.

Finding 2 (gravity, destination WGI): In an OLS gravity model with 184 partners and 9 years (1,496 observations), the destination's lagged WGI composite has a positive association with Peru's exports to that destination (coefficient 0.667, $p < 0.001$, $R^2 = 0.721$). The association survives the two-year lag, the outlier trim, and the median aggregation that were estimated. It does not survive first differences. PPML was not estimated. The coefficient is not an effect of Peruvian governance.

Finding 3 (why the designs differ): The panel and the gravity model do not estimate the same object. The panel uses a competitiveness component in thousands of US dollars and absorbs between-partner differences with fixed effects. The gravity model uses log export levels and gets its WGI slope from differences across destinations. An earlier draft said the panel is five times smaller than a minimum panel size in the literature and attributed that minimum to Riker (2022). Riker (2022) does not state a minimum panel size. The sentence is withdrawn.

Finding 4 (dimensions): Each of the six destination dimensions is positive and significant when entered alone. Regulatory Quality (0.681) and Government Effectiveness (0.671) have the largest coefficients. The ranking describes destinations. It is not evidence on which Peruvian reform would raise exports. An earlier draft attributed that ranking to a working paper that does not exist. That attribution is withdrawn.

Finding 5 (endogeneity): The association is robust to lag extension (t–2: 0.668, $p < 0.001$), outlier removal, and aggregation choices, but it does not survive first differences (–0.063, $p = 0.906$), and there is a significant reverse path from current exports to subsequent governance ($p < 0.001$). The evidence is therefore consistent with a stable between-country levels relationship, not with a within-country causal dynamic. We report this as an association throughout, and do not claim causal identification.

Future research: Three extensions would strengthen the analysis: (i) estimating the gravity model at the HS2 or HS4 product level to identify sector-specific governance effects; (ii) applying a difference-in-differences design exploiting the entry into force of CPTPP for Peru and other members to identify the causal effect of preferential market access; and (iii) obtaining valid instruments for governance — ideally exogenous institutional shocks such as constitutional reforms or international governance assessments — to support a causal interpretation.

References

Abreo, C., R. Bustillo, and C. Rodriguez. 2021. "The Role of Institutional Quality in the International Trade of a Latin American Country: Evidence from Colombian Export Performance." *Journal of Economic Structures* 10: 24. https://doi.org/10.1186/s40008-021-00253-5

Acemoglu, D., and S. Johnson. 2005. "Unbundling Institutions." *Journal of Political Economy* 113(5): 949–995. https://doi.org/10.1086/432166

Álvarez, I. C., J. Barbero, A. Rodríguez-Pose, and J. L. Zofío. 2018. "Does Institutional Quality Matter for Trade? Institutional Conditions in a Sectoral Trade Framework." *World Development* 103: 72–87. https://doi.org/10.1016/j.worlddev.2017.10.010

Anderson, J. E. 1979. "A Theoretical Foundation for the Gravity Equation." *American Economic Review* 69(1): 106–116.

Anderson, J. E., and D. Marcouiller. 2002. "Insecurity and the Pattern of Trade: An Empirical Investigation." *Review of Economics and Statistics* 84(2): 342–352. https://doi.org/10.1162/003465302317411587

Anderson, J. E., and E. van Wincoop. 2003. "Gravity with Gravitas: A Solution to the Border Puzzle." *American Economic Review* 93(1): 170–192. https://doi.org/10.1257/000282803321455214

Cameron, A. C., J. B. Gelbach, and D. L. Miller. 2011. "Robust Inference with Multiway Clustering." *Journal of Business & Economic Statistics* 29(2): 238–249. https://doi.org/10.1198/jbes.2010.07136

de Groot, H. L. F., G.-J. Linders, P. Rietveld, and U. Subramanian. 2004. "The Institutional Determinants of Bilateral Trade Patterns." *Kyklos* 57(1): 103–123. https://doi.org/10.1111/j.0023-5962.2004.00245.x

Kaufmann, D., A. Kraay, and M. Mastruzzi. 2011. "The Worldwide Governance Indicators: Methodology and Analytical Issues." *Hague Journal on the Rule of Law* 3(2): 220–246. https://doi.org/10.1017/S1876404511200046

Leamer, E. E., and R. M. Stern. 1970. *Quantitative International Economics*. Boston: Allyn and Bacon.

Mayer, T., and S. Zignago. 2011. "Notes on CEPII's Distances Measures: The GeoDist Database." CEPII Working Paper 2011-25. http://www.cepii.fr/PDF_PUB/wp/2011/wp2011-25.pdf

Méon, P.-G., and K. Sekkat. 2004. "Does the Quality of Institutions Limit the MENA's Integration in the World Economy?" *The World Economy* 27(9): 1475–1498. https://doi.org/10.1111/j.0378-5920.2004.00661.x

North, D. C. 1990. *Institutions, Institutional Change and Economic Performance*. Cambridge: Cambridge University Press.

Riker, D. 2022. "Regulatory Quality and Trade Performance: An Econometric Analysis." USITC Economics Working Paper 2022-10-A. https://www.usitc.gov/publications/332/working_papers/ecwp_2022-10-a_regulatory_quality.pdf

Santos Silva, J. M. C., and S. Tenreyro. 2006. "The Log of Gravity." *Review of Economics and Statistics* 88(4): 641–658. https://doi.org/10.1162/rest.88.4.641

Tinbergen, J. 1962. *Shaping the World Economy: Suggestions for an International Economic Policy*. New York: Twentieth Century Fund.

Tyszynski, H. 1951. "World Trade in Manufactured Commodities, 1899–1950." *The Manchester School* 19(3): 272–304. https://doi.org/10.1111/j.1467-9957.1951.tb00012.x

Volpe Martincus, C., and J. Carballo. 2008. "Is Export Promotion Effective in Developing Countries? Firm-Level Evidence on the Intensive and the Extensive Margins of Exports." *Journal of International Economics* 76(1): 89–106. https://doi.org/10.1016/j.jinteco.2008.05.002

Wei, S.-J. 2000. "How Taxing Is Corruption on International Investors?" *Review of Economics and Statistics* 82(1): 1–11. https://doi.org/10.1162/003465300558533

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